22 The number of rabbits on a farm at the end of month n is P_n
The number of rabbits at the end of the next month is given by $P_{n+1} = 1.2P_n - 50$

At the end of March there are 200 rabbits on the farm.

(a) Work out how many rabbits there will be on the farm at the end of June.

End March: 200 rabbits.

End April: 1.2 × 200 – 50 = 190.

End May: 1.2 × 190 – 50 = 178.

End June: 1.2 × 178 – 50 = 163.6

≈ 164 rabbits.

3 marks total
April 190
May 178, June 163.6
About 164 rabbits

(3)

(b) Considering your results in part (a), suggest what will happen to the number of rabbits on the farm after a long time.

The values keep falling (200, 190, 178, 163.6).

Eventually the population would die out.

1 mark
Long-term decline

(1)

(Total for Question 22 is 4 marks)

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